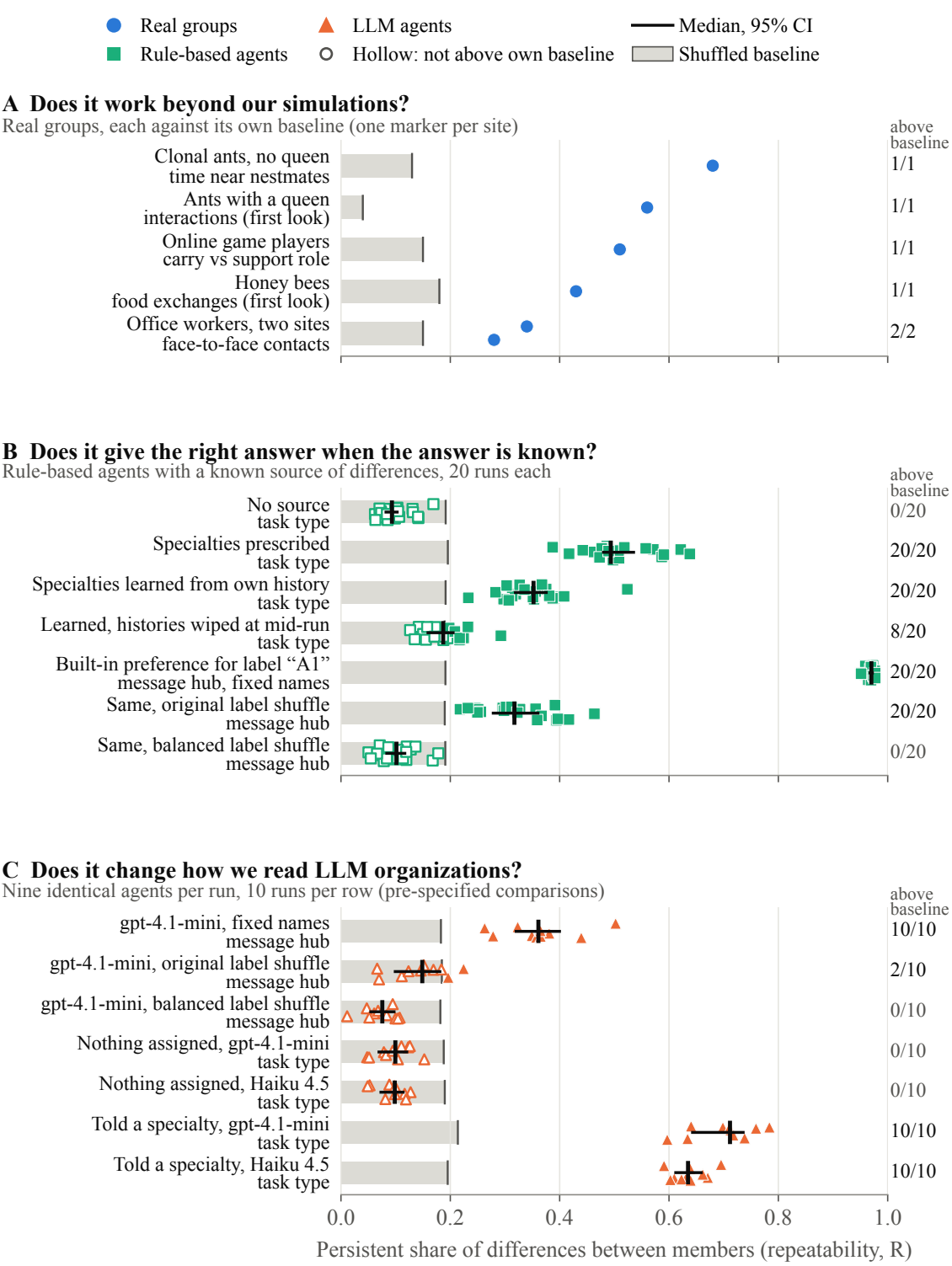

Real groups
Rule-based agents
LLM agents
Hollow: not above own baseline
Median, 95% CI
Shuffled baseline
A Does it work beyond our simulations?
Real groups, each against its own baseline (one marker per site)
above baseline
Clonal ants, no queen time near nestmates
1/1
Ants with a queen interactions (first look)
1/1
Online game players carry vs support role
1/1
Honey bees food exchanges (first look)
1/1
Office workers, two sites face-to-face contacts
2/2
B Does it give the right answer when the answer is known?
Rule-based agents with a known source of differences, 20 runs each
above baseline
No source task type
0/20
Specialties prescribed task type
20/20
Specialties learned from own history task type
20/20
Learned, histories wiped at mid-run task type
8/20
Built-in preference for label "A1" message hub, fixed names
20/20
Same, original label shuffle message hub
20/20
Same, balanced label shuffle message hub
0/20
C Does it change how we read LLM organizations?
Nine identical agents per run, 10 runs per row (pre-specified comparisons)
above baseline
gpt-4.1-mini, fixed names message hub
10/10
gpt-4.1-mini, original label shuffle message hub
2/10
gpt-4.1-mini, balanced label shuffle message hub
0/10
Nothing assigned, gpt-4.1-mini task type
0/10
Nothing assigned, Haiku 4.5 task type
0/10
Told a specialty, gpt-4.1-mini task type
10/10
Told a specialty, Haiku 4.5 task type
10/10
0.0
0.2
0.4
0.6
0.8
1.0
Persistent share of differences between members (repeatability, R)